

33.2 Clustered data and compound symmetry

Whole plots are one instance of a pattern that is everywhere in applied work: observations arrive in groups — households, schools, hospitals, litters, survey clusters — and within a group resemble one another more than they resemble observations from elsewhere. The simplest description gives every pair within a group the same correlation and every pair across groups none. What that does to least squares is sharp and exactly computable: it inflates the variance of some coefficients and *deflates* that of others.

33.2.1 The equicorrelated cluster model

Definition 33.2.1 — Clustered errors with compound symmetry

The n observations fall into G **clusters**, cluster g containing m_g of them, with $n = \sum_g m_g$. The model is $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ with $E \boldsymbol{\varepsilon} = \mathbf{0}$ and

$$\text{Cov}(\boldsymbol{\varepsilon}) = \mathbf{V} = \text{diag}(\boldsymbol{\Sigma}_1, \dots, \boldsymbol{\Sigma}_G), \quad \boldsymbol{\Sigma}_g = \sigma^2 \{(1 - \rho)\mathbf{I}_{m_g} + \rho \mathbf{1}\mathbf{1}^\top\}, \quad (33.2.1)$$

a block diagonal matrix with one block per cluster. This covariance is called **compound symmetry** or **equicorrelation**, and ρ is the **intraclass correlation**. The matrix is positive definite iff $-1/(\max_g m_g - 1) < \rho < 1$. The clusters are **balanced** if $m_g = m$ for all g .

Compound symmetry is what a single shared random effect produces: if $\varepsilon_i = u_{g(i)} + e_i$ with u_g and e_i uncorrelated, of variances σ_u^2 and σ_e^2 , then Eq. 33.2.1 holds with $\sigma^2 = \sigma_u^2 + \sigma_e^2$ and $\rho = \sigma_u^2/\sigma^2$ — the one-way random effects model of Proposition 32.1.1 and Exercise (B1), whose power to inflate an F statistic was met at Exercise (C1). So read, ρ is the fraction of variance attributable to the cluster and is nonnegative; negative intraclass correlations occur — a fixed quota within a cluster, competition among littermates — but not from a random-effect representation.

The results below take the clusters balanced, $m_g = m$; the clean statements fail otherwise (Exercise (B2)), balance being what makes the two strata of Lemma 33.1.1 eigenspaces of $\mathbf{V}$.

33.2.2 Which coefficients suffer

Call $\mathbf{v}$ **cluster-constant** if it is constant on every cluster and **cluster-centred** if its entries sum to zero within every cluster. With $\mathbf{P}_W$ the orthogonal projection replacing each entry by its cluster mean, the two conditions are $\mathbf{P}_W \mathbf{v} = \mathbf{v}$ and $\mathbf{P}_W \mathbf{v} = \mathbf{0}$, and every vector is the sum of one of each.

Proposition 33.2.1 — Compound symmetry and the design effect

Let Eq. 33.2.1 hold with $m_g = m$ for all g , let $\lambda_w = \sigma^2\{1 + (m-1)\rho\}$ and $\lambda_s = \sigma^2(1-\rho)$, and let $\text{rank}(\mathbf{X}) = p$.

- (a) $\mathbf{V} = \lambda_w \mathbf{P}_W + \lambda_s (\mathbf{I} - \mathbf{P}_W)$. Its eigenvalues are λ_w with multiplicity G and λ_s with multiplicity $n - G$, and

$$\mathbf{V}^{-1} = \frac{1}{\lambda_s} \left(\mathbf{I} - \frac{m\rho}{1 + (m-1)\rho} \mathbf{P}_W \right).$$

- (b) $\mathcal{C}(\mathbf{V}\mathbf{X}) \subseteq \mathcal{C}(\mathbf{X})$ iff $\mathcal{C}(\mathbf{P}_W \mathbf{X}) \subseteq \mathcal{C}(\mathbf{X})$. In particular, if $\mathbf{X} = [\mathbf{X}_b, \mathbf{X}_w]$ with every column of $\mathbf{X}_b$ cluster-constant and every column of $\mathbf{X}_w$ cluster-centred, then least squares is generalized least squares and every least squares estimate of an estimable function is best linear unbiased.
- (c) Let $\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}$ be estimable and let $\mathbf{c} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \boldsymbol{\lambda}$, so that $\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}} = \mathbf{c}^\top \mathbf{y}$. If $\mathbf{c}$ is cluster-constant then

$$\text{Var}(\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}) = \{1 + (m-1)\rho\} \cdot \sigma^2 \|\mathbf{c}\|^2,$$

and if $\mathbf{c}$ is cluster-centred then $\text{Var}(\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}) = (1-\rho) \sigma^2 \|\mathbf{c}\|^2$. In both cases $\sigma^2 \|\mathbf{c}\|^2$ is the variance the estimate would have under $\text{Cov}(\boldsymbol{\epsilon}) = \sigma^2 \mathbf{I}$; the multiplier is the **design effect**. In either case $\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}$ is best linear unbiased even if (b) fails.

- (d) Suppose $\mathcal{C}(\mathbf{X}) = \mathcal{S}_b \oplus \mathcal{S}_w$ with $\mathcal{S}_b \subseteq \mathcal{C}(\mathbf{P}_W)$ of dimension p_b and $\mathcal{S}_w \subseteq \mathcal{C}(\mathbf{P}_W)^\perp$ of dimension $p_w = p - p_b$. Then the usual residual mean square has

$$E(s^2) = \frac{\lambda_w(G - p_b) + \lambda_s(n - G - p_w)}{n - p}, \quad (33.2.2)$$

and for any $\mathbf{c}$ as in (c) the usual variance estimate $v = s^2 \boldsymbol{\lambda}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \boldsymbol{\lambda}$ satisfies $E(v) / \text{Var}(\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}) = E(s^2) / \lambda$, with $\lambda = \lambda_w$ or λ_s according to the stratum of $\mathbf{c}$.

Proof.

- (a) Within one cluster, $\mathbf{1}\mathbf{1}^\top = m\bar{\mathbf{J}}_m$, so the block is $\sigma^2\{(1-\rho)\mathbf{I}_m + m\rho\bar{\mathbf{J}}_m\} = \lambda_s(\mathbf{I}_m - \bar{\mathbf{J}}_m) + \lambda_w\bar{\mathbf{J}}_m$, using $\lambda_s + m\rho\sigma^2 = \lambda_w$. Assembling the blocks gives the stated form, since $\mathbf{P}_W$ is block diagonal with blocks $\bar{\mathbf{J}}_m$. A matrix $\lambda_w \mathbf{P} + \lambda_s(\mathbf{I} - \mathbf{P})$ with $\mathbf{P}$ an orthogonal projection of rank G has the two eigenvalues with those multiplicities and inverse $\lambda_w^{-1} \mathbf{P} + \lambda_s^{-1}(\mathbf{I} - \mathbf{P}) = \lambda_s^{-1}\{\mathbf{I} - (1 - \lambda_s/\lambda_w)\mathbf{P}\}$, and $1 - \lambda_s/\lambda_w = m\rho/\{1 + (m-1)\rho\}$.
- (b) By (a), $\mathbf{V}\mathbf{X} = \lambda_s \mathbf{X} + (\lambda_w - \lambda_s)\mathbf{P}_W \mathbf{X}$, so $\mathcal{C}(\mathbf{V}\mathbf{X}) \subseteq \mathcal{C}(\mathbf{X})$ iff $\mathcal{C}(\mathbf{P}_W \mathbf{X}) \subseteq \mathcal{C}(\mathbf{X})$ (the case $\lambda_w = \lambda_s$, that is $\rho = 0$, being trivial). Under the stated partition, $\mathbf{P}_W \mathbf{X}_b = \mathbf{X}_b$ and $\mathbf{P}_W \mathbf{X}_w = \mathbf{0}$, so $\mathcal{C}(\mathbf{P}_W \mathbf{X}) = \mathcal{C}(\mathbf{X}_b) \subseteq \mathcal{C}(\mathbf{X})$. The conclusion is Theorem 6.9.3 together with Corollary 7.2.3, as in Theorem 33.1.2(a); the general statement is Theorem 31.2.2.
- (c) $\text{Var}(\mathbf{c}^\top \mathbf{Y}) = \mathbf{c}^\top \mathbf{V} \mathbf{c}$. If $\mathbf{P}_W \mathbf{c} = \mathbf{c}$ then $\mathbf{V} \mathbf{c} = \lambda_w \mathbf{c}$ and the variance is $\lambda_w \|\mathbf{c}\|^2$; if $\mathbf{P}_W \mathbf{c} = \mathbf{0}$ then it is $\lambda_s \|\mathbf{c}\|^2$. Under $\text{Cov}(\boldsymbol{\epsilon}) = \sigma^2 \mathbf{I}$ it would be $\sigma^2 \|\mathbf{c}\|^2$, and dividing gives the two multipliers. For the last claim, write λ for λ_w or λ_s according to the stratum of $\mathbf{c}$, so that $\mathbf{V} \mathbf{c} = \lambda \mathbf{c}$ in either case, and let $\mathbf{a}^\top \mathbf{Y}$ be any linear unbiased estimate of $\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}$. Then $\mathbf{X}^\top (\mathbf{a} - \mathbf{c}) = \mathbf{0}$, so $\mathbf{a} = \mathbf{c} + \mathbf{d}$ with $\mathbf{d} \perp \mathcal{C}(\mathbf{X})$ and in particular $\mathbf{c}^\top \mathbf{d} = 0$. The cross term $\mathbf{c}^\top \mathbf{V} \mathbf{d} = \lambda \mathbf{c}^\top \mathbf{d}$ vanishes and

$$\mathbf{a}^\top \mathbf{V} \mathbf{a} = \lambda \|\mathbf{c}\|^2 + \mathbf{d}^\top \mathbf{V} \mathbf{d} \geq \lambda \|\mathbf{c}\|^2,$$

with equality iff $\mathbf{d}^\top \mathbf{V} \mathbf{d} = 0$, that is $\mathbf{d} = \mathbf{0}$, $\mathbf{V}$ being positive definite. So $\mathbf{c}^\top \mathbf{Y}$ is the unique best linear unbiased estimate. This is the criterion behind Theorem 31.2.2: a linear estimate $\mathbf{c}^\top \mathbf{Y}$ of an estimable function is best as soon as $\mathbf{V} \mathbf{c} \in \mathcal{C}(\mathbf{X})$.

- (d) By Theorem 19.2.2(b), whose part (b) uses $\mathbf{X}$ only through $\mathbf{M}$ and $\text{rank}(\mathbf{X}) = p$ and so holds verbatim here, $E(s^2) = \text{tr}\{(\mathbf{I} - \mathbf{M})\mathbf{V}\}/(n - p)$ with $\mathbf{M}$ the projection onto $\mathcal{C}(\mathbf{X})$. Now $\text{tr}\{(\mathbf{I} - \mathbf{M})\mathbf{V}\} = \lambda_w \text{tr}\{(\mathbf{I} - \mathbf{M})\mathbf{P}_W\} + \lambda_s \text{tr}\{(\mathbf{I} - \mathbf{M})(\mathbf{I} - \mathbf{P}_W)\}$. Because $\mathbf{M} = \mathbf{P}_{S_b} + \mathbf{P}_{S_w}$ with $\mathbf{P}_{S_b}\mathbf{P}_W = \mathbf{P}_{S_b}$ and $\mathbf{P}_{S_w}\mathbf{P}_W = \mathbf{0}$,

$$\text{tr}\{(\mathbf{I} - \mathbf{M})\mathbf{P}_W\} = \text{tr}(\mathbf{P}_W) - \text{tr}(\mathbf{P}_{S_b}) = G - p_b,$$

and similarly $\text{tr}\{(\mathbf{I} - \mathbf{M})(\mathbf{I} - \mathbf{P}_W)\} = (n - G) - p_w$. This gives Eq. 33.2.2. Finally $v = s^2 \|\mathbf{c}\|^2$ because $\|\mathbf{c}\|^2 = \boldsymbol{\lambda}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{X} (\mathbf{X}^\top \mathbf{X})^{-1} \boldsymbol{\lambda} = \boldsymbol{\lambda}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \boldsymbol{\lambda}$ for estimable $\boldsymbol{\lambda}$, so $E(v)/\text{Var}(\mathbf{c}^\top \mathbf{Y}) = E(s^2) \|\mathbf{c}\|^2 / (\lambda \|\mathbf{c}\|^2)$.

■

The design effect cuts both ways. A regressor measured at the cluster level — the school's curriculum, the village's water supply — is cluster-constant, and its variance is multiplied by $1 + (m - 1)\rho$: with $m = 50$ and $\rho = 0.05$ the factor is 3.45, so 5000 pupils in 100 schools carry the information of about 1450 independent pupils, the **effective sample size** $n/\{1 + (m - 1)\rho\}$. Since the factor grows linearly in m , extra clusters buy far more than extra members per cluster (Exercise (A2)). A cluster-centred regressor is in the opposite position: its design effect $1 - \rho$ is less than one, the shared component cancelling out of within-cluster comparisons — the subplot stratum again.

Idea. Clustering does not simply lose information; it moves it from between-cluster to within-cluster comparisons. A design that can ask its question within clusters gains, one that must ask it between clusters loses, in proportion to the cluster size.

Part (d) says what the usual output does with all this: s^2 averages $\mathbf{V}$ over the residual space while the variance of an estimate reads $\mathbf{V}$ along one direction, so reported and true variance stand in the ratio $E(s^2)/\lambda$, the same for every direction in a stratum.

Example. (Forty clusters of five)

Take $G = 40$ clusters of $m = 5$, with $\rho = 0.4$, $\sigma^2 = 1$, one cluster-level and one within-cluster regressor besides the intercept, so $p_b = 2$ and $p_w = 1$. Then $\lambda_w = 2.6$ and $\lambda_s = 0.6$, and the design effects are exactly 2.60 and 0.60, as the listing confirms from the full 200×200 covariance matrix. By Eq. 33.2.2, $E(s^2) = \{2.6(38) + 0.6(159)\}/197 = 0.9858$, so the reported variance of the cluster-level coefficient is 0.379 times the truth and that of the within-cluster coefficient 1.643 times the truth. A nominal 95% interval for the first is about 1.62 times too short, and the simulation below finds its coverage to be 0.759.

```
import numpy as np

m, rho, sigma2 = 5, 0.4, 1.0                                # cluster size, intraclass
correlation
S = sigma2 * ((1 - rho) * np.eye(m) + rho * np.ones((m, m)))
w = np.linalg.eigvalsh(S)                                    # one large, m-1 equal small
eigenvalues
print("eigenvalues of the cluster covariance:", w.round(4))
```

```
print("large:", sigma2 * (1 + (m - 1) * rho), "  small:", sigma2 * (1 - rho))
```

[Run this code online](#)

```
def design(G, m, seed):
    """Cluster indicator, a cluster-level regressor and a within-cluster regressor.
    """
    rng = np.random.default_rng(seed)
    g = np.repeat(np.arange(G), m)
    xb = np.repeat(rng.normal(0, 1, G), m)          # constant inside each cluster
    xw = rng.normal(0, 1, G * m)
    xw = xw - np.repeat(xw.reshape(G, m).mean(axis=1), m)  # sums to zero inside
    # each cluster
    X = np.column_stack([np.ones(G * m), xb, xw])
    return g, X

G = 40
g, X = design(G, m, 2024)
V = np.kron(np.eye(G), S)          # block diagonal, one block per
    # cluster
XtXinv = np.linalg.inv(X.T @ X)
cov_true = XtXinv @ X.T @ V @ X @ XtXinv          # the real covariance of the
    # OLS estimate
cov_iid = sigma2 * XtXinv          # what the usual formula
    # reports
deff = np.diag(cov_true) / np.diag(cov_iid)
print("design effects (intercept, between, within):", deff.round(4))
print("predicted:", [1 + (m - 1) * rho, 1 + (m - 1) * rho, 1 - rho])
```

[Run this code online](#)

33.2.3 Standard errors that survive the model

Generalized least squares with the right $\mathbf{V}$ fixes everything, and by [Proposition 33.2.1\(b\)](#) is not even needed for the point estimate when the regressors split into cluster-constant and cluster-centred parts. But ρ is rarely known and compound symmetry is itself a guess. With many clusters the alternative is to keep ordinary least squares and estimate its covariance from the clusters themselves: the sandwich idea of [Section 21.4](#) with the cluster in place of the observation. Writing $\mathbf{X}_g$ and $\hat{\boldsymbol{\epsilon}}_g$ for the rows and least squares residuals of cluster g , the **cluster-robust** covariance estimator is, as in Exercise (C1),

$$\hat{\mathbf{U}}_{\text{CR}} = (\mathbf{X}^\top \mathbf{X})^{-1} \left(\sum_{g=1}^G \mathbf{X}_g^\top \hat{\boldsymbol{\epsilon}}_g \hat{\boldsymbol{\epsilon}}_g^\top \mathbf{X}_g \right) (\mathbf{X}^\top \mathbf{X})^{-1}. \quad (33.2.3)$$

Nothing about $\mathbf{V}$ is assumed except that different clusters are uncorrelated: within a cluster the covariance may be arbitrary, and may differ from cluster to cluster — far weaker than [Eq. 33.2.1](#).

Remark (Consistency of the cluster-robust estimator). Let the clusters be independent, cluster g contributing rows $\mathbf{X}_g$ and errors $\boldsymbol{\epsilon}_g$ of mean zero and covariance $\boldsymbol{\Sigma}_g$, with p fixed, and assume (K1) $m_g \leq m_{\max}$ for all g , and $\mathbb{E}\|\boldsymbol{\epsilon}_g\|^4 \leq K$; (K2) $\|\mathbf{x}_{(i)}\| \leq C$ for all i ; (K3) $n^{-1} \mathbf{X}^\top \mathbf{X} \rightarrow \mathbf{A}$, positive definite. Write $\mathbf{U} = \text{Cov}(\hat{\boldsymbol{\beta}}) = (\mathbf{X}^\top \mathbf{X})^{-1} (\sum_g \mathbf{X}_g^\top \boldsymbol{\Sigma}_g \mathbf{X}_g) (\mathbf{X}^\top \mathbf{X})^{-1}$.

Then $n(\hat{\mathbf{U}}_{\text{CR}} - \mathbf{U}) \rightarrow \mathbf{0}$ in probability as $G \rightarrow \infty$; and if also $n^{-1} \sum_g \mathbf{X}_g^\top \boldsymbol{\Sigma}_g \mathbf{X}_g \rightarrow \mathbf{B}$, positive definite, then $\sqrt{n}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}) \rightarrow N_p(\mathbf{0}, \mathbf{A}^{-1} \mathbf{B} \mathbf{A}^{-1})$ and the Wald statistic of a hypothesis $\mathbf{A}^\top \boldsymbol{\beta}$ of rank q , formed with $\hat{\mathbf{U}}_{\text{CR}}$, tends to $\chi^2(q)$.

The proof is that of Theorem 21.4.2 with the cluster in place of the observation: the cluster score $\mathbf{s}_g = \mathbf{X}_g^\top \boldsymbol{\varepsilon}_g$ replaces $\mathbf{x}_{(i)} e_i$, every sum has G independent terms instead of n , and each bound goes through with $m_{\max} C$ in place of C (Exercise (C1)).

Note where the asymptotics live: $G \rightarrow \infty$, with cluster sizes bounded; the number of *observations* is irrelevant. A study of a million patients in eight hospitals has eight independent pieces of information about anything varying at the hospital level, and no covariance estimator manufactures more.

```
def cluster_sandwich(X, resid, g):
    """The cluster-robust covariance estimate: one outer product per cluster."""
    XtXinv = np.linalg.inv(X.T @ X)
    meat = np.zeros((X.shape[1], X.shape[1]))
    for k in np.unique(g):
        s = X[g == k].T @ resid[g == k]                # the cluster's score
        contribution                                     # contribution
        meat += np.outer(s, s)
    return XtXinv @ meat @ XtXinv

rng_one = np.random.default_rng(55)
n, p = X.shape
e_one = (np.linalg.cholesky(S) @ rng_one.standard_normal((m, G))).T.reshape(-1)
y_one = X[:, 1] + e_one                                # the between coefficient is 1
b_one = XtXinv @ X.T @ y_one
res_one = y_one - X @ b_one
s2_one = res_one @ res_one / (n - p)
se_naive_one = np.sqrt(s2_one * XtXinv[1, 1])
se_cr_one = np.sqrt(cluster_sandwich(X, res_one, g)[1, 1] * G / (G - 1))
se_true_one = np.sqrt(cov_true[1, 1])
print(f"between coefficient {b_one[1]:.4f}: naive se {se_naive_one:.4f}, "
      f"cluster-robust {se_cr_one:.4f}, true sd {se_true_one:.4f}")
```

[Run this code online](#)

On the data generated here the cluster-level coefficient is 1.0376 against a true value of 1, with a usual standard error of 0.0856, a cluster-robust one of 0.1600 (the listing scales the estimator by the simple factor $G/(G - 1)$) and an exact standard deviation of 0.1239: the naive interval is far too short, the robust one conservative on this sample.

Figure 33.2.1 shows the pattern over repeated samples, with the fuller correction $G/(G - 1) \cdot (n - 1)/(n - p)$. The naive interval covers about three quarters of the time — 0.757 to 0.781 across the six designs — whatever the number of clusters, its error being a bias rather than a variance. The cluster-robust interval is badly short with 8 clusters (coverage 0.874), still short with 20 (0.917) and acceptable by 160 (0.951); generalized least squares with the correct covariance is right throughout.

Warning. With few clusters the cluster-robust standard error is not a safe default: its bias is downward and substantial, and referring the statistic to $t(G - 1)$, the usual remedy and the one used in the figure, is not enough. Better are a bias correction of the residuals, a bootstrap over whole clusters (Exercise (C1)), or a covariance model fitted by restricted maximum likelihood, which does not need G large. Twenty clusters is a

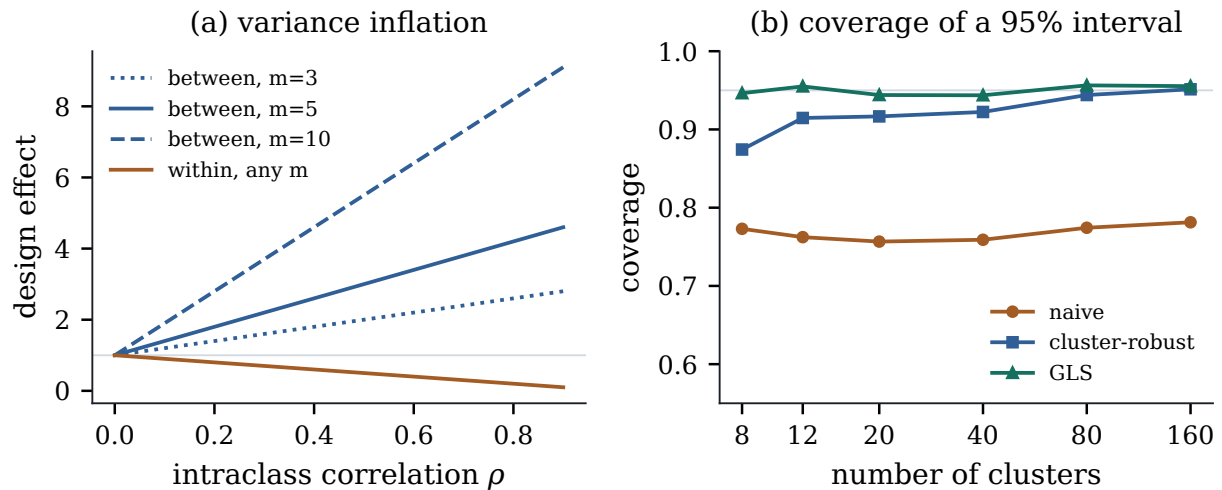


Figure 33.2.1: (a) The design effect of Proposition 33.2.1(c): inflation $1 + (m - 1)\rho$ for a cluster-level regressor, deflation $1 - \rho$ for a within-cluster one. (b) Simulated coverage of nominal 95% intervals for a cluster-level coefficient, with $m = 5$ and $\rho = 0.4$.

common rule of thumb; it is a floor, not a target.

33.2.4 Exercises

33.2.4.1 A. Check your understanding

Exercise A1.

Show from Proposition 33.2.1(a) that the compound symmetric matrix is positive definite iff $-1/(m - 1) < \rho < 1$, and that the random-effects representation $\varepsilon_i = u_{g(i)} + e_i$ forces $\rho \geq 0$. Give a mechanism that produces $\rho < 0$.

Exercise A2.

A trial can afford $n = 1200$ pupils and must randomize schools, not pupils, to the two arms. The intraclass correlation of the outcome is $\rho = 0.10$. Compare the effective sample size for 12 schools of 100 pupils, 40 schools of 30, and 120 schools of 10.

Solution. The effective sample size is $n/\{1 + (m - 1)\rho\}$: $1200/10.9 = 110$, $1200/3.9 = 308$ and $1200/1.9 = 632$ pupils. As m grows the effective size tends to $n/(m\rho) = G/\rho$, which depends only on the number of schools. For a school-level treatment the number of schools is nearly the whole story, and pupils within a school are cheap but nearly redundant.

33.2.4.2 B. Practice

Exercise B1.

Let $\mathbf{X} = [\mathbf{X}_b, \mathbf{X}_w]$ as in Proposition 33.2.1(b), with $\mathbf{1} \in \mathcal{C}(\mathbf{X}_b)$. Show that $\mathbf{M} = \mathbf{M}_b + \mathbf{M}_w$ with $\mathbf{M}_b$ and $\mathbf{M}_w$ the projections onto $\mathcal{C}(\mathbf{X}_b)$ and $\mathcal{C}(\mathbf{X}_w)$, that the residual sum of squares

splits into a between-cluster and a within-cluster part, and that the two corresponding mean squares are unbiased for λ_w and λ_s .

Solution. Any column of $\mathbf{X}_b$ is fixed by $\mathbf{P}_W$ and any column of $\mathbf{X}_w$ is annihilated by it, so $\mathcal{C}(\mathbf{X}_b) \perp \mathcal{C}(\mathbf{X}_w)$ and Theorem 6.3.8 gives $\mathbf{M} = \mathbf{M}_b + \mathbf{M}_w$. Then $\mathbf{I} - \mathbf{M} = (\mathbf{P}_W - \mathbf{M}_b) + (\mathbf{I} - \mathbf{P}_W - \mathbf{M}_w)$, a sum of two orthogonal projections of ranks $G - p_b$ and $n - G - p_w$. By Proposition 33.2.1(a), $\mathbf{V}$ acts as λ_w on the range of the first and λ_s on the range of the second, so Theorem 2.3.1 gives the expectations $\lambda_w(G - p_b)$ and $\lambda_s(n - G - p_w)$. Dividing by the ranks gives the unbiased estimates, of which Eq. 33.2.2 is the pooled average.

Exercise B2.

Let the clusters have unequal sizes m_g . Show that $\mathbf{V}$ of Eq. 33.2.1 has eigenvalues $\sigma^2\{1 + (m_g - 1)\rho\}$, one per cluster, together with $\sigma^2(1 - \rho)$ with multiplicity $n - G$. Deduce that $\mathbf{1}$ is an eigenvector only if all clusters have the same size, so that the intercept alone breaks Proposition 33.2.1(b).

33.2.4.3 C. Going deeper

Exercise C1.

Carry out in detail the three steps of the cluster-sandwich argument above, with the constants: show that (K1)–(K3) give $\|\mathbf{X}_g\| \leq m_{\max}C$ and $E\|\mathbf{X}_g^\top \boldsymbol{\varepsilon}_g\|^2 \leq m_{\max}^2 C^2 K^{1/2}$; that $\text{Var}(\mathbf{a}^\top \mathbf{X}_g^\top (\boldsymbol{\varepsilon}_g \boldsymbol{\varepsilon}_g^\top - \boldsymbol{\Sigma}_g) \mathbf{X}_g \mathbf{a})$ is bounded uniformly in g ; and that $n^{-1} \sum_g \|\mathbf{t}_g\|^2 \leq m_{\max}^2 C^2 (\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})^\top \mathbf{A}_n (\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})$. Where exactly does the number of clusters, rather than of observations, enter?

Exercise C2.

Show that $E(\mathbf{X}_g^\top \hat{\boldsymbol{\varepsilon}}_g \hat{\boldsymbol{\varepsilon}}_g^\top \mathbf{X}_g) \neq \mathbf{X}_g^\top \boldsymbol{\Sigma}_g \mathbf{X}_g$ in general, by writing $\hat{\boldsymbol{\varepsilon}}_g = \boldsymbol{\varepsilon}_g - \mathbf{X}_g(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \boldsymbol{\varepsilon}$ and expanding, and show that the leading correction is of order p/G . Why is the bias downward, and what does this suggest about the factor $G/(G-1) \cdot (n-1)/(n-p)$ used in the coverage simulation behind Figure 33.2.1?